

Khushali Pariyal

AI Engineer | Agentic AI Engineer | RAG & Generative AI | LLMOps | Decision Intelligence

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years of experience designing, building, and deploying production **Agentic AI** and **Generative AI** systems: multi-agent workflows with **LangGraph**, **tool calling**, and **Model Context Protocol (MCP)**, **RAG pipelines** spanning vector search, multimodal document processing, LLM inference, and automated model evaluation, and **LLMOps**. Skilled in AI agents on **AWS Bedrock**, prompt engineering, computer vision, and MLOps, and in building AI decision-intelligence systems. Built **Meadow**, a human-in-the-loop supervised coding agent, with a record of shipping systems that cut manual effort by up to 80% and improve accuracy and time-to-market.

TECHNICAL SKILLS

Agentic AI: Agentic AI, AI Agents, Multi-Agent Systems, LangGraph, LangChain, Supervisor Agents, Autonomous Agents, Tool Calling, Function Calling, Model Context Protocol (MCP), Agent Orchestration, Agent Harness Engineering, Human-in-the-Loop, Context Engineering

Generative AI & RAG: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Embeddings, Semantic Search, Knowledge Graphs, Multimodal AI, Vision-Language Models (VLM), OCR, NLP, Model Evaluation, Guardrails

AI Decision Systems: AI Decision Intelligence, Investment Analytics, Predictive Analytics, Price Intelligence, Recommendation Systems, Scoring & Ranking Models, Explainable AI, Conversational AI

ML / Computer Vision: Machine Learning, Deep Learning, Computer Vision, PyTorch, YOLO, CNN, DeepLabV3, PatchCore, Stable Diffusion

Cloud & Infrastructure: AWS Bedrock, AgentCore, Amazon Textract, OpenSearch, FAISS, EventBridge, Lambda, Serverless, AWS, Azure, Ollama

MLOps / DevOps: MLOps, LLMOps, CI/CD, Docker, Git, GitLab, GitHub Actions, Apache Kafka

Languages / Frameworks: Python, TypeScript, JavaScript, Java, C, C++, SQL, Node.js, Django, Spring Boot, React

Databases / APIs: MySQL, MongoDB, PostgreSQL, Redis, REST APIs, Telegram Bot API

WORK EXPERIENCE

Tata Consultancy Services (TCS) – Gandhinagar, Gujarat

AI Engineer (Systems Engineer)

Aug 2024 – Present

- Architected an autonomous **Agentic AI** system that turns stakeholder requirements into shipped software: full architecture flow, **code generation**, automated **code review**, and deployment, using **AWS Bedrock**, **AgentCore**, and a **RAG** pipeline (Amazon Textract OCR, OpenSearch vector database, EventBridge, Lambda) with event-driven orchestration, reducing feature **time-to-market by 75%**.
- Engineered an end-to-end **Generative AI RAG pipeline** that generates ADAS test cases from NCAP specification documents, combining multimodal ingestion (text, tables, images), chunking, vector embeddings, **FAISS** semantic retrieval, LLM inference, and agentic validation loops to reduce hallucination; cut manual test creation by **70%** and doubled test coverage.
- Built deep learning and **computer vision** models for railway inspection using **DeepLabV3** (segmentation) and **PatchCore** (anomaly detection) across multi-camera pipelines processing 20K+ images per cycle; applied **Stable Diffusion** for synthetic data, improving precision/recall by **18–22%** under variable lighting and weather.
- Deployed an LLM-powered developer tooling platform using **DeepSeek** models for automated code review, static analysis, architectural validation, and security vulnerability detection in **CI/CD** pipelines, cutting manual review effort by **80%** and raising pre-merge defect detection.

Project Intern (TCS)

Jan 2024 – May 2024

- Automated an end-to-end **MLOps** pipeline on **Azure** with **GitLab CI/CD** to train, version, and deploy a **YOLO**-based computer vision model for real-time traffic sign and speed-limit detection, achieving ~30 FPS inference and >90% detection accuracy.

Infolabz – Web Development Intern

Jul 2023 – Aug 2023

- Engineered a **Django**-powered RESTful News API for real-time article aggregation across multiple categories with scalable backend architecture.

Microsoft – Microsoft Engage Mentee

May 2022 – Jun 2022

- Developed a scalable movie **recommendation system** using content-based filtering, collaborative filtering, and ranking algorithms under mentorship from Microsoft engineers.

PROJECTS

MEADOW

Agentic AI, MCP, Tool Calling, Harness Engineering, TypeScript

- Built Meadow, an open-source, local-first **agentic system** that turns a request (text, voice, or PLAN.md) into finished, tested code: it writes a spec and phased plan, then drives a coding engine (Cursor CLI) phase by phase after **human-in-the-loop** approval via dashboard or Telegram.
- Engineered the **agent harness**: each phase runs on its own git branch with guard checks and tests, failures return to the engine with the real error for up to 3 fix attempts, and only passing work merges; added headless-browser flow tests with desktop and mobile screenshots.
- Implemented a **supervisor agent** (Ollama, FreeLLMAPI, or Claude) that reviews failures, reports progress over Telegram, and runs independent phases in parallel; integrated **tool calling** and **MCP** tools with approval gates, audit logs, and a CodeAtlas knowledge graph for cited answers.

CASORA

AI Decision Intelligence, LLM, Semantic Search, Investment Analytics, Full-Stack

- Built and deployed **Casora**, an AI-powered real estate platform to search, compare, and understand homes across India using **verified listings** and neighbourhood-level **price intelligence**, replacing broker-driven discovery with data-backed decisions.
- Designed an **AI investment decision layer** that turns market data (price trends, comparable listings, predictive analytics) into explainable buy/invest recommendations via scoring and ranking models; applied **LLMs**, **semantic search**, and **prompt engineering** for grounded conversational discovery.

SPLITMATE

OCR, Vision-Language Models (VLM), Gemini API, LLM, Full-Stack

- Built and deployed an AI bill-splitting web app using **OCR** and **Vision-Language Models (VLMs)** to extract line items, quantities, taxes, and service charges from receipt images and PDFs into structured data, with a **human-in-the-loop** review workflow.
- Designed a deterministic math engine so the **LLM** never performs arithmetic (totals, proportional tax, rounding reconciled) and a debt-graph algorithm that minimizes group settlements; supports configurable LLM endpoints (Bring Your Own AI) with **Gemini**.

EDUCATION

Government Engineering College Gandhinagar – B.E. in Computer Engineering

Government Polytechnic for Girls Ahmedabad – Diploma in Information Technology

Aug 2021 – Jun 2024 | CGPA: 8.37

Aug 2018 – Jul 2021 | CGPA: 9.69